



An α -vector predictive value iteration algorithm for transportation infrastructure maintenance under partially observable conditions

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ABSTRACT

Transportation infrastructure is degrading over time and poses the risk of failure when exposed to a dynamic environment. Periodic inspection is often implemented to assess the requirement of maintenance. However, the inspected conditions can only partially reflect the underlying degradation, complicating the decision of maintenance. Moreover, inspections of early degradation often have no value-adding to condition improvement and incur a portion of unnecessary expenses. To address the abovementioned issues, we propose a sequential predictive maintenance policy that accounts for the partial observation of the infrastructure's condition to reduce unnecessary inspections. The schedule of inspection timings is predicted according to the estimated Remaining Useful Life distribution, adaptive to stochastic degradation. We demonstrate that the optimal value function is piecewise linear and convex when decision epochs are non-periodic. Leveraging this insight, we have designed an α -vector Predictive Value Iteration algorithm (α -PVI) to optimize the transportation infrastructure maintenance policy. The α -PVI algorithm further reduces the time complexity compared with the Point-Based Value Iteration algorithm. Our designed approach is verified through an application for the maintenance optimization of pavements and bridges. The results demonstrate that the α -PVI algorithm reduces unnecessary inspection costs by on average 61.25% when compared to the periodic inspection approach. The α -PVI algorithm constructs a new paradigm of predictive maintenance under partially observable conditions.

1. Introduction

The transportation infrastructure is irreplaceable in contemporary society and daily activities. Their normal operation provides foundational services, facilitating economic growth, and enabling the exchange of information. However, when exposed to a dynamic environment, they naturally degrade and pose the risk of failure. Based on the 2021 ASCE infrastructure report card [1], bridges, as a critical component in transportation infrastructure, are currently in a poor state with visible deficiencies, suggesting a heightened likelihood of failures. Inspection and maintenance are effective means to mitigate degradation and prevent failure. The United States Department of Transportation is facing about a 50% funding gap to maintain the operation of the transportation infrastructure. Hence, a cost-effective inspection and maintenance policy for transportation infrastructure is essential.

The designed service life of transportation infrastructure, such as bridges, is often as long as 50 years [2]. The degradation is stochastic when exposed to a dynamic environment and is often modeled as a Markov model to optimize the maintenance policy. Mandiartha et al. utilized a Markov chain to analyze the maintenance of road

network pavement according to historical roughness progression and costs [3]. Lethanh et al. incorporated mechanistic-empirical models of chloride-induced corrosion to assess the transition probabilities of degradation [4]. Connor et al. formulated the corrosion initiation and nonlinear degradation of chloride-induced degradation as Markov models to optimize the maintenance plan [5]. Anoop et al. optimized the maintenance of reinforced concrete bridges by a Markov model with fuzzy states [6]. Markov models are suitable for formulating the degradation of Multi-State Systems and optimizing the maintenance of transportation infrastructure. In real-world applications, the fundamental assumption of Markov models—specifically, the notion of perfect inspection—is often compromised. This occurs because the actual state of the system is typically partially revealed by the inspection signals. Take bridges for example, many signals, such as loss of bar diameter and concrete crack width [7], can be only utilized indirectly to assess the underlying degradation state. The partially observable conditions should be incorporated into maintenance optimization to enhance accuracy.

To handle this scenario, the Partially Observable Markov Decision Process (POMDP) is designed to optimize the maintenance with

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Abbreviations

MDP	Markov Decision Process
HMM	Hidden Markov Model
POMDP	Partially Observable Markov Decision Process
POMSS	Partially Observable Multi-State Systems
RUL	Remaining Useful Life
α -PVI	α -vector Predictive Value Iteration algorithm
PBVI	Point-Based Value Iteration algorithm

Notations

T	Finite horizon
t_d	Predictive inspection timing
S	State set
$\mathcal{O}$	Signal set
A	Action set
$P_{ij}(t_{d+1} - t_d)$	Transition probability from the state i to state j between $t_{d+1} - t_d$
E_{ij}	Emission probability that the signal is state j when underlying degradation is state i
a_1	No action
a_2	Minor maintenance
a_3	Major maintenance
a_4	Replacement
a_I	Inspection
c_1	Cost of a_1
c_2	Cost of a_2
c_3	Cost of a_3
c_4	Cost of a_4
c_I	Cost of a_I
c_F	Deviation cost rate after failure
c_L	Deviation cost rate before failure
τ	Decision variable of next inspection interval
$B_i^{O_{0:t_d}}(t_d)$	Belief when the underlying degradation is state i at predictive inspection timing t_d
$P_{ij}^{a_l}$	Effect of a_l from state i to state j
$F^{a_l}(t O_{0:t_d})$	RUL distribution when the recorded signals are $O_{0:t_d}$ and taken action is a_l
F_j	RUL distribution of state j
R_T	Reliability threshold
$V(B^{O_{0:t_d}}(t_d))$	Value function of belief vector $B^{O_{0:t_d}}(t_d)$
$\alpha_i(t_d)$	α -vector when the underlying degradation is i at predictive inspection timing t_d

partially observable conditions. The POMDP extends a Markov Decision Process (MDP) [8] by synthesizing observation uncertainty. It formulates the partial observation as a probability mapping between the inspected signals and underlying degradation states. Song et al. incorporated a POMDP with the value of information analysis to illustrate the degradation for non-stationary systems [9]. Morato et al. incorporated dynamic Bayesian networks with POMDPs to optimize inspection and maintenance for a structural component under fatigue degradation [10]. Nguyen et al. utilized a POMDP to optimize inspection and maintenance policies based on inspected signals [11]. Compare et al. formulated a POMDP to optimize the operation and maintenance policy for gas transmission networks under the demand variation [12]. Liu et al. employed a POMDP to tackle the optimization problem of multi-type inspections and derived structural properties for the optimal policy [13].

Another important model to optimize maintenance under partially observable conditions is the Hidden Markov model (HMM). Duan et al. introduced an adaptive inspection mode to predict failures in systems characterized by hidden degradation, employing a multivariate continuous-time HMM [14]. Duan proposed a dynamic monitoring and detection approach for partially observable machines under discrete

multivariate observations [15]. Arcieri et al. proposed an approach to assessing parameters of the POMDP based on recorded data by sampling from an HMM and applied the approach to maintenance management for railway tracks [16]. Cheng et al. jointly optimized inspection and condition-based mission abort policy for a safety-critical system using an HMM and derived structural properties for the control limit [17]. Guo and Liang integrated the POMDP and HMM to reduce unnecessary inspections and optimize predictive maintenance under partial observation [18]. The POMDP and HMM effectively integrate stochastic degradation processes with partial observation, representing well-established and robust approaches in the field of maintenance policy and decision-making. Often, they necessitate periodic inspections to gather the degradation signals for informed maintenance decisions. However, in practical scenarios, such as within the transportation infrastructure, the implementation of periodic inspections frequently results in no condition improvement during the early degradation, as the systems often remain in satisfactory operational states, thus requiring no maintenance. Consequently, these unnecessary inspections can lead to inefficiencies and increased costs, which indicates the potential to improve.

To enhance the efficiency of inspection and maintenance methods, various scholars have investigated irregular inspection techniques and predictive maintenance policies. Cheng et al. developed dynamic, sample-based sequential inspection and maintenance schemes for products at risk of multiple failure modes during storage [19]. Hu and Chen proposed a Weibull proportional hazards model that combines degradation and failure time data, allowing for the calculation of a precise distribution for Remaining Useful Life (RUL) at each inspection point and the creation of an optimal maintenance plan [20]. Huynh et al. devised a novel predictive maintenance framework for k -out-of- n : F systems that endure continuous deterioration, considering both stochastic dependencies from shared environmental factors and economic dependencies due to the allocation of setup costs [21]. Wang et al. suggested a dual-tiered inspection policy model for single-component industrial systems, structured around a three-stage failure dynamic [22].

To our knowledge, the research about non-periodic inspection and predictive maintenance of transportation infrastructure considering partial observable conditions is still underexplored. Hence, we introduce an innovative sequential predictive maintenance policy that schedules inspections by estimating the RUL under conditions of partial observability, specifically for transportation infrastructure systems. Our approach predicts inspection timings based on the RUL and determines maintenance actions, such as minor and major maintenance, by a sequential decision-making framework. Inspection schedules are dynamically updated in response to the progression of system degradation, aiming to minimize redundant inspections. Multiple evaluations from predictive inspections enhance the understanding of the system's hidden degradation process, thereby supporting timely and effective maintenance decisions. A Bayesian method is applied to historical partially observable data to estimate the underlying degradation state. Thereafter, the RUL is predicted by accounting for all possibilities of underlying degradation states. Moreover, we illustrate that the optimal value function is piecewise linear and convex when decision epochs are non-periodic. Leveraging this insight, we design a new α -vector Predictive Value Iteration algorithm (α -PVI) among non-periodic decision epochs, which is well-suited for managing maintenance policies in Partially Observable Multi-State Systems (POMSS) exhibiting continuous belief state spaces. The designed algorithm reduces the time complexity compared with the Point-Based Value Iteration algorithm (PBVI) and constructs a new paradigm of predictive maintenance. The α -PVI algorithm has been applied to practical degradation scenarios encountered in pavements and bridges. Pavements and bridges are studied systems in this paper. These types of transportation infrastructure play an important role in providing essential services for people. However, along with their operation, they are degrading naturally and require huge maintenance costs. Currently, routine inspection and condition-based maintenance are the main means to keep the reliability of these systems, which indicates the potential to reduce costs. Compared with routine inspection, non-periodic inspection based on degradation can reduce unnecessary inspections. For these types of systems, many observed signals, such as loss of bar diameter and concrete crack width, are utilized indirectly to assess the underlying degradation states. Hence, partial observation is incorporated into our approach. Three contributions can be summarized as follows.

1. *Novel Sequential Predictive Maintenance under Partial Observability:* A novel sequential predictive maintenance policy is proposed by predicting inspection timings based on the RUL under partially observable conditions for transportation infrastructure.
2. *Reducing Computational Complexity:* The α -PVI algorithm further reduces the time complexity compared to the PBVI algorithm.
3. *Cost-efficiency in inspections:* The proposed maintenance policy reduces unnecessary inspection costs compared to the periodic inspection approach by 61.25% on average.

Compared with the previous work [18], this paper introduces an innovative algorithm termed α -PVI algorithm developed to enhance maintenance policies for the transportation infrastructure. Distinguished by its superior computational efficiency and enhanced interpretability, the α -PVI algorithm directly computes optimal maintenance policies across the entire continuous belief state space without necessitating discretization. We present propositions 2, 3, 4, and 5 to support our algorithm's efficacy. Propositions 2 and 3 demonstrate the closed form of the optimal inspection interval and the piecewise linear and convex of the optimal value function, laying the theoretical groundwork for the designed algorithm. Proposition 4 demonstrates that the optimal policy belongs to a type of control-limit policy, thus improving practical applicability, while Proposition 5 highlights our algorithm's computational superiority.

The remaining paper is structured as follows. Section 2 presents the formulation of the POMSS. Section 3 designs an α -PVI algorithm for the POMSS with continuous belief state space. Section 4 presents the case study of maintenance optimization for pavements and bridges to illustrate the advantage of the α -PVI algorithm compared with the periodic inspection approach. Section 5 concludes the whole paper.

2. Partially Observable Multi-State Systems formulation

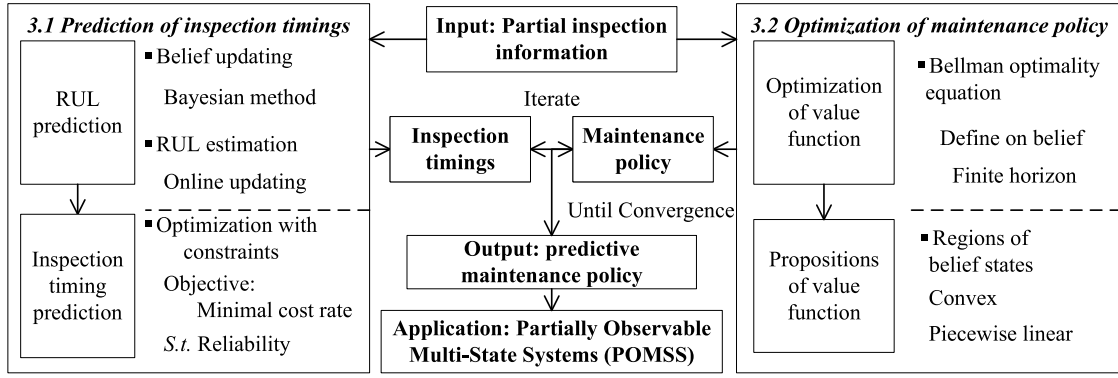
Transportation infrastructures are modeled as Partially Observable Multi-State Systems (POMSS). The real degradation state is partially observable by inspections in our model formulation. The optimization horizon is set as T . In the finite horizon, we set several inspections to obtain the signals of the degradation. The inspection timings in the α -PVI algorithm are predicted based on the RUL, indicating that the inspection timings are non-periodic. This is the essential difference from the POMDP with periodic inspection. The inspection timings are expressed as $\{t_1, t_2, \dots, t_d, \dots, t_N = T\}$. We assume the degradation of the systems follows a Markov process with a state set as $S = \{1, 2, \dots, i, \dots, n\}$. The state transition probability between the underlying degradation states is denoted as $P_{ij}(t_{d+1} - t_d) = P(S_{t_{d+1}} = j | S_{t_d} = i)$. The derivation of state transition probability in non-periodic decision epochs can be expressed as Eq. (1) based on the Chapman-Kolmogorov equation.

$$P_{ij}(t_{d+1} - t_d) = \int_0^{t_{d+1} - t_d} \sum_k P(S_{t_d + \Delta} = k | S_{t_d} = i) P_{kj}[t_{d+1} - (t_d + \Delta)] d\Delta \quad (1)$$

where $P(S_{t_d + \Delta} = k | S_{t_d} = i)$ represents the state transition probability among the unit-time Δ . Along with the degradation of the systems, we need to assess their conditions through inspections to inform and support maintenance decisions. Often the failure state n is self-revealing. Failure indicates the system service is destroyed, which requires immediate maintenance. Other states, apart from the state n , are estimated through inspection. The signals that can be inspected are denoted as $\mathcal{O} = \{1, 2, \dots, j, \dots, m\}$. The emission probability between underlying degradation states and signals is as $E_{jk} = P(O_{t_d} = k | S_{t_d} = j)$.

The actions optimized in our model include inspection and maintenance. Inspection (a_I) obtains the signals for helping estimate underlying degradation. The cost is expressed as c_I . Then we decide whether to implement the maintenance. The alternative actions can be divided into four types:

1. No action (a_1): it has no influence on degradation with the cost c_1 .
2. Minor maintenance (a_2): it can restore the state to a better state with the cost c_2 . The state transition probability is often set based on expert knowledge or historical records.
3. Major maintenance (a_3): The effect of a_3 is better than a_2 for restoration but require more costs. The cost of a_3 is set as c_3 . The best condition is to restore as new, which has the same effect as active replacement.

Fig. 1. The structure of the α -PVI algorithm.

4. Passive replacement (a_4): it restores the systems as new with the cost c_4 . In our model, we consider stochastic degradation and partial observation, which indicates the possibility of unexpected failure between successive inspections. Once a failure occurs, the service quality of transportation infrastructure will be affected. Hence, we set a_4 to be passively implemented immediately when an unexpected failure happens between two successive inspections.

The order of c_1, c_2, c_3 , and c_4 is often set as $c_1 = 0 < c_2 < c_3 < c_4$, where c_4 includes the downtime loss. a_4 is passively taken place. Hence, we only need to optimize the first three actions in active maintenance optimization. a_1, a_2 , and a_3 form the action set A . Compared to the long horizon, the duration of maintenance is negligible in our model. The state-action transition probability can be expressed as

$$P_{ij}^{a_l}(t_{d+1} - t_d) = P(S_{t_{d+1}} = j | S_{t_d} = i, a_l) \\ = \sum_h P(S_{t_d}^+ = h | S_{t_d} = i, a_l) P(S_{t_{d+1}} = j | S_{t_d}^+ = h) \quad (2)$$

where $a_l \in \{a_1, a_2, a_3\}$. $P(S_{t_d}^+ = h | S_{t_d} = i, a_l)$ represents the effect of a_l . The main assumptions in our formulation are listed below.

1. The degradation of the system follows a Markov model.
2. The duration of maintenance is negligible in our model.

The α -PVI algorithm features an adaptive predictive schedule for non-periodic inspections, designed to accommodate stochastic degradation. After the formulation of the POMSS, we present the design process of the α -PVI algorithm in the following section.

3. Algorithm design

α -PVI algorithm is a newly designed algorithm for sequential predictive maintenance under partially observable conditions. The resulting predictive inspection policy based on estimated RUL distribution can reduce unnecessary inspections adaptive to stochastic processes. The structure of the α -PVI algorithm is demonstrated in Fig. 1. Based on partial inspection data, the proposed algorithm predicts inspection schedules utilizing estimated RUL distributions. Through the propositions of the optimal value function derived based on the Bellman optimality equation, the algorithm optimizes the maintenance policy among a continuous belief state space. Iteratively refining over a finite horizon until convergence, this approach yields a predictive maintenance policy suitable for applying to the POMSS. Then we present the design process of the α -PVI algorithm.

3.1. Prediction of inspection timings

In the α -PVI algorithm, the inspection timings are predicted based on the RUL adaptive to stochastic degradation, which can effectively

reduce unnecessary inspection when compared with periodic inspection. Due to partial observation, we need to estimate the underlying degradation for RUL prediction. Different from the MDP, the underlying degradation cannot be obtained completely by inspection. We utilize the belief vectors to represent the estimation of the underlying degradation based on inspected signals. The belief is defined as $B_i^{O_{0:t_d}}(t_d) = P(S_{t_d} = i | O_0, O_{t_1}, \dots, O_{t_d})$. For compact, we utilize $O_{0:t_d}$ to represent $O_0, O_{t_1}, \dots, O_{t_d}$. All $B_i^{O_{0:t_d}}(t_d) \forall i$ form the belief vector at the inspection timing t_d . The belief is derived based on the Bayesian approach as

$$B_j^{O_{0:t_d},k}(t_{d+1}) = \frac{\sum_i B_i^{O_{0:t_d}}(t_d) P_{ij}^{a_l}(t_{d+1} - t_d) E_{jk}}{\sum_j \sum_i B_i^{O_{0:t_d}}(t_d) P_{ij}^{a_l}(t_{d+1} - t_d) E_{jk}} \quad (3)$$

where k represents the possible signal at timing t_{d+1} .

Proposition 1. Given action a_l and inspected signals $O_{0:t_d}$, the RUL distribution can be expressed as

$$F^{a_l}(t | O_{0:t_d}) = \sum_{i=1}^n F_j(t_d) B_i^{O_{0:t_d}}(t_d) P_{ij}^{a_l} \quad (4)$$

where $F_j(t_d)$ represents the RUL distribution when the current state is j .

The proof of Proposition 1 can be found in Appendix A. Proposition 1 indicates the estimation of RUL prediction under partially observable conditions. Then we apply the concept of deviation cost to determine the next inspection timing by balancing the cost of maintenance and risk of failure.

Proposition 2. Given the RUL distribution $F^{a_l}(t | O_{0:t_d})$, the inspection timing t_{d+1} is predicted by economic objective as Eq. (5) subject to the reliability threshold as Eq. (6) with deviation cost c_F, c_L .

$$t_{d+1} = t_d + \arg \min_{\tau} \frac{\int_0^{\tau} c_F F^{a_l}(t | O_{0:t_d}) dt + \int_{\tau}^{\infty} c_L \bar{F}^{a_l}(t | O_{0:t_d}) dt}{\int_0^{\tau} \bar{F}^{a_l}(t | O_{0:t_d}) dt} \quad (5)$$

$$\bar{F}^{a_l}(t | O_{0:t_d}) \geq R_T \quad (6)$$

Then the optimal inspection interval τ satisfies $\tau = \min\{\tau_1, \tau_2\}$, where τ_1 satisfies

$$\int_0^{\tau_1} \frac{F^{a_l}(t | O_{0:t_d}) - F^{a_l}(t | O_{0:t_d})}{\bar{F}^{a_l}(\tau_1 | O_{0:t_d})} dt = \frac{c_L}{c_F} \mu \quad (7)$$

τ_2 satisfies

$$\bar{F}^{a_l}(\tau_2 | O_{0:t_d}) = R_T \quad (8)$$

The proof of Proposition 2 can be found in Appendix B. Eq. (5) optimizes the next inspection timing by comparing the risk of failure

and maintenance cost. Eq. (6) sets the reliability threshold R_T . According to the Eq. (5) and (6), we can obtain the optimal inspection interval τ^* under reliability threshold. μ represents the mean of the RUL distribution. The scheduling of inspections is predicted based on the newest updated RUL estimation. Consequently, the derived inspection intervals are non-periodic and adaptive to stochastic degradation. Following the acquisition of these predictive inspection schedules, we proceed to discuss the optimization of maintenance policies.

3.2. Optimization of maintenance policy

After obtaining the predictive inspection timings, the maintenance is optimized based on the Bellman optimality equation at non-periodic decision epochs. Different from the MDP, the value function is defined on belief vectors rather than states. The value function is expressed as Eq. (9).

$$\begin{aligned} V(\mathbf{B}^{O:t_d}(t_d)) \\ = \min_{a_l \in A} \left[\sum_{i=1}^{i=n} B_i^{O:t_d}(t_d) \sum_{j=1}^{j=n} P_{ij}^{a_l}(t_{d+1} - t_d) \right. \\ \left. \times \sum_k E_{jk} \left\{ c_I + c_l + V(\mathbf{B}^{O:t_d,k}(t_{d+1})) \right\} \right] \end{aligned} \quad (9)$$

Different from the POMDP and MDP with periodic decision epochs, the α -PVI algorithm is optimized among non-periodic decision epochs. To compact the derivation, we set

$$q_i^{a_l} = \sum_{j,k} P_{ij}^{a_l}(t_{d+1} - t_d) E_{jk} (c_I + c_l) \quad (10)$$

Then the value function can be expressed as Eq. (11).

$$\begin{aligned} V(\mathbf{B}^{O:t_d}(t_d)) = \min_{a_l \in A} \left[\sum_{i=1}^{i=n} B_i^{O:t_d}(t_d) q_i^{a_l} + \right. \\ \left. \sum_{i,j,k} B_i^{O:t_d}(t_d) P_{ij}^a(t_{d+1} - t_d) E_{jk} V(\mathbf{B}^{O:t_d,k}(t_{d+1})) \right] \end{aligned} \quad (11)$$

The initial condition is

$$V(\mathbf{B}^{O:0:T}(T)) = \sum_i B_i^{O:0:T}(T) q_i(T) = \mathbf{B}^{O:0:T}(T) \mathbf{q}(T) \quad (12)$$

where $q_i(T)$ represents the depreciated value of the systems. The belief states are continuous, which indicates that the computation of the value function for the belief vectors is huge. To simplify the computation, we derive Proposition 3 of the value function.

Proposition 3. *The optimal value function of the α -PVI algorithm is piecewise linear and convex with non-periodic decision epochs and can be expressed as*

$$V(\mathbf{B}^{O:t_d}(t_d)) = \min_{\mathcal{L}} \left[\sum_{i=1}^{i=n} \alpha_i^{\mathcal{L}}(t_d) B_i^{O:t_d}(t_d) \right], t_d < T \quad (13)$$

where $\mathcal{L}$ is a function about a_l and signal $O_{t_{d+1}}$. $\alpha_i^{\mathcal{L}}(t_d)$ is recurrently updated with Eq. (14).

$$\alpha_i^{\mathcal{L}}(t_d) = q_i^{a_l} + \sum_{k,j} P_{ij}^{a_l}(t_{d+1} - t_d) E_{jk} \alpha_j^{\mathcal{L}}(t_{d+1}), t_d < T \quad (14)$$

The initial $\alpha_i^{\mathcal{L}}(T)$ is $q_i(T)$.

The proof of Proposition 3 can be referred to Appendix C. In other words, the iteration of the value function can be transformed into a computation of the α -vectors. α -vectors contain the information of the optimal maintenance policy. For a continuous belief state space, α -vectors can divide it into several regions. Each region represents one type of optimal maintenance action. Through this format, we can

reduce the computation of the value function. Proposition 3 is also suitable for the POMDP with periodic decision epochs.

Proposition 4. *The optimal maintenance policy derived from α -PVI algorithm follows a control limit policy with respect to belief vectors.*

For a detailed proof of this proposition, please refer to Appendix D. Then we present the details of the α -PVI algorithm in the following subsection.

3.3. α -Vector predictive value iteration algorithm design

Value iteration and policy iteration algorithms are classic algorithms for the MDP. However, when the scenario extends to partial observation, the belief states are continuous and infinite, which will lead to computation being intractable for the value iteration algorithm and policy iteration algorithm. In practice, huge computation costs limit the application of the POMDP. To avoid huge computation costs caused by continuous belief states, we design an α -vector Predictive Value Iteration algorithm (α -PVI). When the decision epochs are periodic, the designed algorithm is also suitable for the POMDP. The α -PVI algorithm is derived from Proposition 3. In this algorithm, the α -vector is iterative at each decision epoch to obtain an optimal maintenance policy. The pseudocode is shown in Algorithm 1.

Algorithm 1 α -PVI algorithm

Input: action cost c_I , $c_l = 0$, c_2 , c_3 , depreciation value $\mathbf{q}(T)$, initial belief vector $\mathbf{B}^{O_0}(0)$, transition probability $P_{ij}^{a_l}(t)$ for $\forall i, j, a_l$, emission $E_{jk} \forall j, k$, RUL distribution $F_i(t)$, deviation cost rate c_F , c_L

Output: predictive inspection timings $t_1^*, t_2^*, \dots, T$, maintenance policy π^*

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1: Initialization: maintenance policy  $\pi$ 
2: repeat
3:   for  $t_d = 0, \dots, t_{N-1}$  do
4:     compute  $t_{d+1}$  based on Proposition 2
5:     compute  $B_j^{O:t_d,k}(t_{d+1})$  based on Eq. (3)
6:   end for // prediction of inspection timings
7:   for  $t_d = t_{N-1}, \dots, 0$  do
8:     compute  $\alpha^{\mathcal{L}}(t_d)$  based on Eq. (14)
9:     prune unnecessary  $\alpha$ 
10:    compute  $V(\mathbf{B}^{O:t_d}(t_d))$  based on Eq. (13)
11:    obtain optimal maintenance policy  $\pi$ 
12:  end for // optimization of maintenance policy
13: until  $\pi$  converged

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The algorithm can be divided into two steps. First, we predict the next inspection timing based on Proposition 2 given the maintenance policy. Second, the maintenance is optimized at predictive inspection timings. If the optimal maintenance policy converges, the joint optimal inspection and maintenance policy can be attained. If not, the inspection timings need to be re-optimized until convergence. The α -vector includes the information about the maintenance policy. Each iteration will create $|A||M|$ types of α -vectors. It seems the computation is still huge. In practice, not all the vectors could provide information on the optimal maintenance policy. Here we provide two rules to prune unnecessary vectors for simplification.

1. If $\alpha_i^{k_1}(t_d) \geq \alpha_i^{k_2}(t_d)$ for $\forall i$, then the $\alpha^{k_1}(t_d)$ is unnecessary.
2. If $\exists \alpha_i^{k_1}(t_d) \leq \alpha_i^{k_2}(t_d)$ for $\forall k_2$, then the $\alpha^{k_1}(t_d)$ is necessary.

The parameters k_1 and k_2 represent two possible observations. Different observations may lead to different α -vectors. Unnecessary vectors should be pruned to reduce the computation.

Proposition 5. *The time complexity of α -PVI algorithm is $|B|^2|M|/|A||N||D|$.*

Table 1
Time complexities of the PBVI algorithm and α -PVI algorithm.

Algorithm	Time complexity
PBVI	$ B ^2 M A N T/A $ [24,25]
α -PVI	$ B ^2 M A N D $

The D represents the number of predictive decision epochs. The PBVI algorithm is a relatively new approximation approach that is currently considered state-of-the-art in the field of POMDPs [23]. The time complexities of α -PVI and PBVI are shown in Table 1. In the α -PVI algorithm, the inspection timings are predicted and can be adaptive to the degradation, which leads to a reasonable reduction when compared with periodic inspection. Hence, the time complexity of the α -PVI is smaller compared to that of the PBVI algorithm.

A discussion about the framework of the α -PVI algorithm is as follows. The inspection interval and maintenance affect each other, which requires an iterative process to optimize them sequentially. Upon determination of the inspection intervals, the maintenance policy can be optimized leveraging the Bellman optimality equation. Conversely, once the maintenance policy is established, the inspection interval can be refined through the application of Eq. (5)–(8), iteratively adjusting until joint optimality. The POMDP enables optimizing the maintenance under equidistant inspection intervals without predicting inspection timings. Compared with the POMDP, the α -PVI algorithm predicts the inspection interval while optimizing maintenance, which is a more versatile approach.

The α -PVI algorithm is suitable for predictive maintenance optimization of the POMSS with continuous belief state space. It iteratively computes α -vectors rather than the value function of belief vectors directly, reducing the computation costs.

4. Case study

Transportation infrastructure is essential for daily living and societal functions, serving as the backbone for the movement of goods and individuals. It is subject to degradation over time, which inherently poses a risk of system failure. The failure of transportation infrastructure may cause serious consequences. Significant financial resources are allocated annually to ensure normal operation. Pavements and bridges are the main elements in transportation infrastructure that require essential inspection and maintenance to ensure reliability. Current practice [26] often involves periodic inspections at biennial intervals to inspect degradation and provide maintenance recommendations. However, this periodic inspection scheme may precipitate unnecessary evaluations during the initial phases of degradation, when the systems may still not necessitate maintenance, thereby engendering inefficient utilization of inspection resources. As for current inspection technologies, the real degradation of transportation infrastructure is partially reflected by inspected signals. To save inspection costs under partially observable conditions, we utilize the designed α -PVI algorithm to optimize the predictive maintenance policy for pavements and bridges.

4.1. Optimization of pavement maintenance

This case study applies the α -PVI algorithm to optimize the maintenance of pavements subject to degradation. The degradation of pavements is divided into five states according to the International Roughness Index (IRI) as demonstrated in Table 2 [27]. The underlying degradation state is hardly observed completely by current inspection technologies. Hence, the emission probabilities are assumed to be normally distributed with zero mean and Standard Deviation (SD) of 0.32 m/km as shown in Table 3 [28]. The Normal distribution is general for maintenance models. The state-action transition probabilities are tabulated in Table 4 [27]. Different actions have different effects on

Table 2
Relation between International Roughness Index (IRI) and Pavement Serviceability Rating (PSR).

State	PSR	IRI (m/km)
1 (very good)	≥ 4	< 0.94
2 (good)	3.5–3.9	0.94–1.48
3 (fair)	3.1–3.4	1.49–1.87
4 (mediocre)	2.6–3.0	1.88–2.68
5 (poor)	≤ 2.5	> 2.68

Table 3
Emission probability limited by inspection technologies for pavements.

$S_{i_t} \backslash O_{i_t}$	1	2	3	4	5
1	0.6	0.4	0	0	0
2	0.2	0.6	0.2	0	0
3	0	0.2	0.6	0.2	0
4	0	0	0.4	0.6	0
5	0	0	0	0	1

degradation. The parameters in Tables 3 and 4 are based on previous expert knowledge from the *Structures Asset Management Planning Toolkit* [29] and Refs. [27,28]. They can be refined if more inspection data is obtained. The corresponding cost parameters are set in Table 5 [30]. The finite horizon is set as 30 years. To determine the next inspection timing, the reliability threshold, as defined in Eq. (8), must be met. This threshold is typically a predefined target value that the infrastructure aims to achieve through maintenance. Different values influence the frequency of inspections. In the case study, we set the target reliability threshold to 99% based on [31].

After the parameter settings, we optimize the inspection and maintenance policy based on the designed α -PVI algorithm. The optimal maintenance policy at each decision epoch is illustrated in Fig. 2.

Apart from the failure state, the other four states of pavements are partially inspected. We utilize a regular tetrahedron to represent the whole belief state space, whose four planes indicate states 1, 2, 3, and 4, respectively. The distance from the inside point to the four planes of the regular tetrahedron represents the belief probability of the first four states. Hence, Fig. 2 represents the optimal maintenance policy of the whole belief state space at predictive inspection timings. At the end of the finite horizon (30th year), the optimal maintenance policy is a_1 . Because the value function of the end finite horizon contains no future costs. The predictive inspection timings are shown below the regular tetrahedra. At the penultimate decision epoch (28.77th year), the state space is divided into two regions. The optimal maintenance of the belief states with high probability in state 1 is a_1 . The optimal maintenance policy of other belief states is a_2 . For the other five decision epochs, the belief states are divided into three regions. The purple line divides the belief state space into two regions, where the optimal maintenance is a_1 and a_2 , and the blue line divides the belief state space into two regions, where the optimal maintenance is a_2 and a_3 . In summary, the optimal maintenance of the belief states with high probability in state 1 is a_1 . The optimal maintenance of the belief states with high probability in state 4 is a_3 . The remaining region represents the optimal maintenance policy a_2 of other belief states. The resulting optimal maintenance policy is consistent with Proposition 3.

Table 6 records the resulting α -vector in maintenance optimization. They are obtained based on Eq. (14). The *italic* α -vectors are optimal and contain the information of the optimal maintenance policy. By combining the current belief vector iteratively updated based on Eq. (3), we can obtain the optimal action. Fig. 3 illustrates the optimal predictive maintenance policy when the initial system is new. The belief vector of states 1–5 is demonstrated in a color bar from top to bottom. Along with the degradation of systems, the belief vectors are changeable as shown in colored rectangles, and the inspection

Table 4
The state-action transition probabilities for pavements.

Action	a_1					a_2					a_3						
S_{t_d}	$S_{t_{d+1}}$					S_{t_d}	$S_{t_{d+1}}$					S_{t_d}	$S_{t_{d+1}}$				
	1	2	3	4	5		1	2	3	4	5		1	2	3	4	5
1	0.84	0.121	0.039	0	0	1	0.8148	0.141	0.0421	0.0021	0	1	0.84	0.121	0.039	0	0
2	0	0.788	0.142	0.07	0	2	0.7140	0.1974	0.0714	0.0142	0.003	2	0.798	0.1543	0.0442	0.0035	0
3	0	0	0.708	0.192	0.1	3	0.3780	0.3697	0.1593	0.0684	0.0246	3	0.672	0.2544	0.0596	0.014	0
4	0	0	0	0.578	0.422	4	0	0.3546	0.3471	0.1777	0.1206	4	0.588	0.2817	0.0982	0.0271	0.005
5	0	0	0	0	1	5	0	0	0	0	1	5	0	0	0	0	1

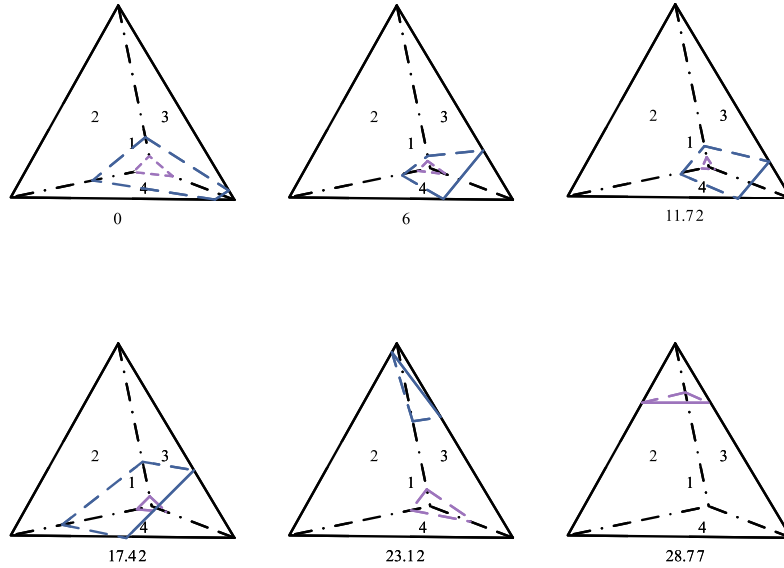


Fig. 2. Optimal maintenance policy at each decision epoch for pavements.

Table 5
The cost parameters settings for pavements.

Action	a_1	a_2	a_3	a_4
Cost	\$0.027/m ²	0	\$16/m ²	\$68/m ²

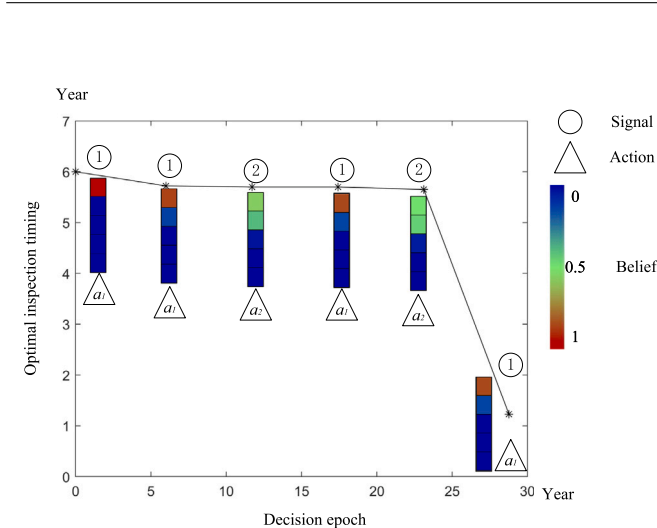


Fig. 3. Predictive maintenance policy when the initial state of the pavement is new.

timings are scheduled flexibly based on the RUL prediction, adapted to stochastic degradation in the form of a line with *. The estimated RUL is affected by the signal in the form of a circle and action in the form of a triangle in Fig. 3.

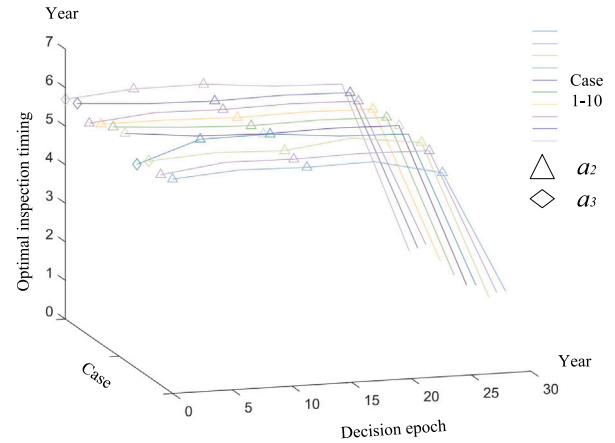


Fig. 4. Optimal inspection and maintenance policies under different initial belief vectors.

To test the adaptation of our designed approach, we apply it under different initial belief vectors as illustrated in Table 7. The optimal inspection and maintenance policies are demonstrated in Fig. 4. Different initial belief vectors will lead to different degradation. Correspondingly, the predictive inspection timings and the optimal maintenance policies are different. Our approach is suitable for different degradation conditions.

The setting of the reliability threshold will affect the results of predictive inspection timings. More specifically, it will affect the results of Eq. (8). To further elucidate the reliability threshold, we explore the sensitivity analysis of the reliability threshold in Table 8. Along with

Table 6
 α -vector in maintenance optimization for pavements.

Decision epoch	Optimal action	α -vector				
0	a_1	3400	8315	15 824	21 086	26 908
		3766	4480	7606	13 204	26 908
		4331	4577	5314	6181	26 908
		1926	4134	9931	16 780	23 774
6	a_1	2211	2593	4435	8445	23 774
		2857	2968	3299	3809	23 774
		1222	4216	9828	14 533	20 390
		1531	2022	4071	8101	20 390
11.72	a_2	2153	2302	2752	3332	20 390
		435	1452	5026	10 197	16 803
		684	900	1904	4369	16 803
		1366	1416	1569	1849	16 803
17.42	a_1	48	573	3018	6901	13 006
		282	417	1039	2702	13 006
		979	1004	1084	1253	13 006
		0.4	0	27	1262	8981
23.12	a_2	219	220	260	419	8981
		931	931	931	933	8981
		0	0	0	0	4518
		0	0	0	0	4518

Table 7
The settings of different initial belief vectors.

Case	Initial belief vector				
1	0.2	0.8	0	0	0
2	0	1	0	0	0
3	0	0	1	0	0
4	0	0	0	1	0
5	0.9	0.1	0	0	0
6	0.8	0.2	0	0	0
7	0.5	0.5	0	0	0
8	0.1	0.9	0	0	0
9	0.3	0.3	0.4	0	0
10	0.25	0.25	0.25	0.25	0

Table 8
The sensitive analysis of reliability thresholds.

R_T	Optimal inspection timing (year)
99.99%	3
99%	7
95%	11
90%	14

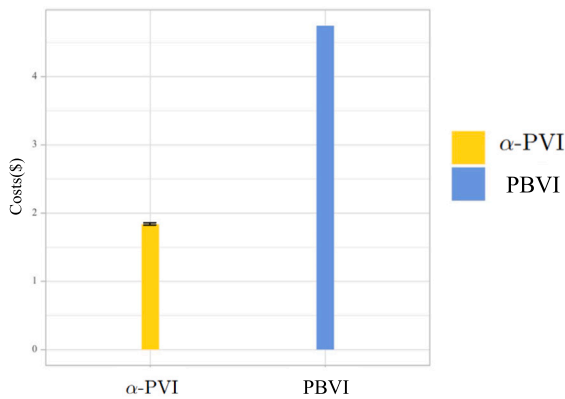


Fig. 5. The comparison of inspection costs between the α -PVI and PBVI algorithm.

the increase in reliability thresholds, the optimal inspection timings when the system is new are decreasing. Intuitively, more frequent inspections can ensure higher reliability. Hence, for specific systems, a suitable reliability threshold is necessary.

To demonstrate the advantage of the α -PVI algorithm, we compare its performance with the PBVI algorithm under 10 different initial belief

Table 9
The comparison of total costs between α -PVI and PBVI algorithms.

Case	Cost(\$)		Reduction
	α -PVI	PBVI	
1	591.26	597.66	1.07%
2	591.67	597.66	1.00%
3	1302.84	1309.75	0.53%
4	1336.84	1344.75	0.59%
5	589.96	597.66	1.29%
6	590.11	597.66	1.26%
7	590.70	597.66	1.16%
8	591.48	597.66	1.03%
9	1302.84	1309.75	0.53%
10	1334.84	1344.75	0.74%
Mean	882.25	889.49	0.92%

Table 10
The inspection intervals of the degradation-state-based policy.

Degradation state	1	2	3	4	5
Optimal inspection interval (year)	7	4	2	0.5	0

vectors. The PBVI algorithm follows the same assumptions, parameters, and settings of the α -PVI algorithm. Fig. 5 illustrates the inspection costs of the PBVI and α -PVI algorithms under 10 cases. For different cases, the α -PVI algorithm predicts inspection timings adapted to degradation. Hence, the inspection costs are different. The PBVI algorithm is implemented under periodic inspections. The costs are the same for these 10 cases. The average reduction is 61.25%. The total costs of inspection and maintenance between α -PVI and PBVI algorithms are listed in Table 9. α -PVI algorithm can reduce 0.92% costs compared with the PBVI algorithm. In the α -PVI algorithm, the inspection timings are predicted and adapted to the degradation while satisfying the reliability threshold. Periodic inspection in the PBVI algorithm is preset and wastes unnecessary costs.

The inspection schedule generated by the α -PVI algorithm exhibits a non-periodic pattern. To highlight the strengths of the proposed method, we conduct a comparative analysis against a degradation-state-based inspection policy. Typically, this type of policy determines the timing of subsequent inspections solely based on the current degradation level, assuming perfect observation conditions. As a result, it produces deterministic inspection intervals when degradation is fully observable. Using identical parameter settings for both approaches, Table 10 presents the inspection intervals derived from the degradation-state-based policy.

Table 11

The predictive inspection interval of α -PVI algorithm and the Degradation-state-based Policy (DP).

Case	Belief vector						Observation		Inspection interval (year)	
									α -PVI	DP
1	0.2	0.8	0	0	0	0	1		4.6	7
2	0.4	0.6	0	0	0	0	1		5.2	7
3	0.3	0.7	0	0	0	0	1		4.9	7
4	0.9	0.1	0	0	0	0	1		6.7	7
5	0.8	0.2	0	0	0	0	1		6.4	7
6	0.2	0.8	0	0	0	0	2		4.6	4
7	0.4	0.6	0	0	0	0	2		5.2	4
8	0.3	0.7	0	0	0	0	2		4.9	4
9	0.9	0.1	0	0	0	0	2		6.7	4
10	0.8	0.2	0	0	0	0	2		6.4	4

Table 12

The comparison between α -PVI algorithm and the Degradation-state-based Policy (DP).

Case	$\geq R_T$		Cost-saving	
	α -PVI	DP	α -PVI	DP
1	✓	×	–	–
2	✓	×	–	–
3	✓	×	–	–
4	✓	×	–	–
5	✓	×	–	–
6	✓	✓	✓	×
7	✓	✓	✓	×
8	✓	✓	✓	×
9	✓	✓	✓	×
10	✓	✓	✓	×

When system conditions are partially observable, inspection intervals should be determined by evaluating the inferred degradation state rather than relying solely on direct observations. Since observable signals may not provide an accurate representation of the true degradation states, inspection policies based solely on observable degradation levels may fail to achieve the desired reliability threshold or cost efficiency. To illustrate this, we examine ten scenarios featuring varying belief vectors and observations. Table 11 displays the predicted inspection intervals produced by both the degradation-state-based policy and the α -PVI algorithm.

The α -PVI algorithm dynamically adjusts inspection intervals based on the estimated underlying degradation states under partially observable conditions, whereas the degradation-state-based policy schedules inspections according to observable degradation levels. To evaluate the performance of both approaches, we consider two key metrics: reliability and cost-saving compliance. The first metric assesses whether the reliability threshold (R_T) is satisfied. The second metric, cost-saving, reflects the reduction in inspection expenses achieved through extended intervals when discounting is applied. Our goal is to obtain a cost-effective inspection policy while meeting the reliability threshold. The cost-saving metric is evaluated only when the reliability threshold is fulfilled. Table 12 presents the comparative results, where a check mark (✓) indicates compliance with the criterion, and a cross (×) denotes non-compliance. – represents the second metric requires no consideration because we have obtained the optimal policy based on the first metric. Across multiple scenarios, the α -PVI algorithm consistently achieves lower costs while adhering to the specified reliability threshold.

To demonstrate the computation efficiency of the α -PVI algorithm, we record the computation time of the α -PVI algorithm and the PBVI algorithm under 10 cases. The results illustrate that the designed α -PVI algorithm performs better than the PBVI algorithm with 57.6% efficiency on average as listed in Table 13. α -PVI algorithm reduces the unnecessary decision epochs by predicting the inspection timings, consequently reducing the computation.

Table 13

The computational time between α -PVI and PBVI algorithms.

Case	Computational time (s)		Efficiency
	α -PVI	PBVI	
1	18.8087	43.5382	56.8%
2	18.6885	46.1665	59.5%
3	18.5459	43.9811	57.8%
4	37.1598	86.1746	56.9%
5	18.3121	43.4565	57.9%
6	19.4849	44.5575	56.3%
7	19.3268	43.3187	55.4%
8	18.4253	44.7125	58.8%
9	17.9512	44.4791	59.6%
10	36.1774	85.0930	57.5%
Mean	22.2881	52.5478	57.6%

Table 14

Emission probability of bridges.

S_{i_d}	O_{i_d}								
	1	2	3	4	5	6	7	8	9
1	0.8	0.15	0.05	0	0	0	0	0	0
2	0.15	0.65	0.15	0.05	0	0	0	0	0
3	0.05	0.15	0.60	0.15	0.05	0	0	0	0
4	0	0.05	0.15	0.60	0.15	0.05	0	0	0
5	0	0	0.05	0.15	0.60	0.15	0.05	0	0
6	0	0	0	0.05	0.15	0.60	0.15	0.05	0
7	0	0	0	0	0.05	0.15	0.65	0.15	0
8	0	0	0	0	0	0	0.05	0.95	0
9	0	0	0	0	0	0	0	0	1

4.2. Optimization of bridge maintenance

Bridges play an important role in transportation infrastructures. As the Introduction describes, the periodic inspections have the potential to be improved. To reduce the portion of unnecessary inspections, we apply our designed approach to bridge maintenance to demonstrate the effectiveness of the α -PVI algorithm. The degradation of bridges is usually divided into nine states. The emission probabilities restricted by inspection technologies are assumed to be normally distributed with zero mean and SD of 0.32m/km as shown in Table 14. The state-action transition probabilities are tabulated in Table 15. The cost parameters are shown in Table 16. We optimize the inspection and maintenance policy within the finite horizon of 60 years.

After the setting of parameters, we optimize the inspection and maintenance policy for bridges as demonstrated in Fig. 6. Table 17 records the resulting optimal α -vector (*italic*) in maintenance optimization. Along with degradation, the optimal action includes a_2 twice and a_1 fourth as shown in Fig. 6 and Table 17. We compare the costs of inspection and maintenance between the α -PVI and PBVI algorithms as illustrated in Fig. 7. The results demonstrate that the α -PVI algorithm can effectively reduce unnecessary inspection costs. The maintenance costs are nearly the same because we implement the optimal maintenance policy in two approaches.

Four remarks can be summarized as follows.

1. As demonstrated in Fig. 2, Fig. 3, and Fig. 6, the α -PVI algorithm is capable of optimizing inspection and maintenance policies for pavements and bridges under partially observable conditions.
2. As illustrated in Fig. 4, the α -PVI algorithm enables optimizing inspection and maintenance policies regardless of the initial conditions of systems effectively.
3. As illustrated in Fig. 5, the α -PVI algorithm schedules predictive inspection adaptive to stochastic degradation and reduces unnecessary inspection costs compared to periodic inspections in the PBVI algorithm by 61.25% on average.
4. As listed in Table 13, the designed α -PVI algorithm performs better than the PBVI algorithm with 57.6% efficiency on average.

Table 15
State-action transition probability of bridges.

	S_{t_d}	$S_{t_{d+1}}$								
		1	2	3	4	5	6	7	8	9
$P(S_{t_{d+1}} S_{t_d}, a_1)$	1	0.97	0.03	0	0	0	0	0	0	0
	2	0	0.85	0.12	0.03	0	0	0	0	0
	3	0	0	0.40	0.46	0.14	0	0	0	0
	4	0	0	0	0.40	0.46	0.14	0	0	0
	5	0	0	0	0	0.40	0.46	0.14	0	0
	6	0	0	0	0	0	0.40	0.46	0.14	0
	7	0	0	0	0	0	0	0.40	0.46	0.14
	8	0	0	0	0	0	0	0	0.40	0.60
	9	0	0	0	0	0	0	0	0	1
$P(S_{t_{d+1}} S_{t_d}, a_2)$	1	0.97	0.03	0	0	0	0	0	0	0
	2	0.83	0.13	0.02	0.02	0	0	0	0	0
	3	0.39	0.39	0.10	0.08	0.03	0.01	0	0	0
	4	0	0.34	0.23	0.27	0.13	0.03	0	0	0
	5	0	0	0.16	0.37	0.31	0.13	0.03	0	0
	6	0	0	0	0.16	0.37	0.31	0.13	0.03	0
	7	0	0	0	0	0.16	0.37	0.31	0.13	0.03
	8	0	0	0	0	0	0.16	0.37	0.32	0.15
	9	0	0	0	0	0	0	0	0	1
$P(S_{t_{d+1}} S_{t_d}, a_3)$	1	0.97	0.03	0	0	0	0	0	0	0
	2	0.97	0.03	0	0	0	0	0	0	0
	3	0.97	0.03	0	0	0	0	0	0	0
	4	0.97	0.03	0	0	0	0	0	0	0
	5	0.97	0.03	0	0	0	0	0	0	0
	6	0.97	0.03	0	0	0	0	0	0	0
	7	0.97	0.03	0	0	0	0	0	0	0
	8	0.97	0.03	0	0	0	0	0	0	0
	9	0.97	0.03	0	0	0	0	0	0	0

Table 16
The cost parameters settings for bridges [32,33].

Action	a_I	a_1	a_2	a_3	a_4
Cost (\$/m ²)	13.5	0	10800	32400	71550

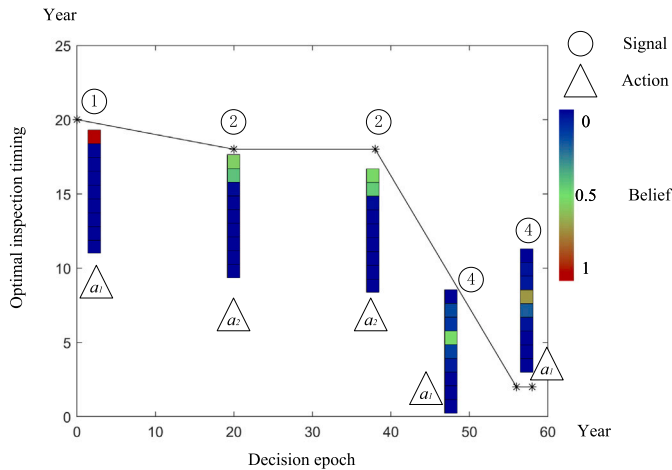


Fig. 6. Predictive maintenance policy when the initial state of the bridge is new.

5. Conclusion

In this paper, a novel α -PVI algorithm is designed to optimize the sequential predictive maintenance policy for transportation infrastructures under partially observable conditions. It can effectively reduce unnecessary inspection costs when compared with periodic inspections. We illustrate that the optimal value function is piecewise linear and convex when decision epochs are non-periodic. The α -PVI algorithm further reduces the time complexity in contrast to the PBVI algorithm when the decision epochs increase. In the case study, the α -PVI algorithm can reduce 0.92% total costs (including 61.25% inspection

costs) on average than the PBVI algorithm while satisfying the reliability threshold with increasing efficiency of 57.6%. It promotes a new paradigm of predictive maintenance and broadens the maintenance approaches for transportation infrastructure under partially observable conditions.

The designed α -PVI algorithm belongs to a type of predictive sequential decision-making approach. When the degradation states are large or the application scenario is the infinite horizon, the computation will increase and may even be unaffordable for real applications. Hence, we will explore more effective algorithms in the future.

CRedit authorship contribution statement

Chunhui Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhenglin Liang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The proof of Proposition 1

In the α -PVI algorithm, we consider the partially observable scenario. Hence, we need to estimate the underlying degradation according to signals. The belief is derived as Eq. (3) iteratively. The action

Table 17
 α -vector in maintenance optimization for bridges.

Decision epoch	Optimal action	α -vector								
0	a_1	49 164	155 850	217 760	223 270	226 170	227 330	227 650	227 710	299 260
		63 164	77 824	133 430	204 710	232 330	235 990	237 710	238 330	299 260
		81 564	81 564	81 564	81 564	81 564	81 564	81 564	81 564	120 700
		24 930	119 120	190 490	198 750	203 580	205 750	206 430	206 570	278 120
20	a_2	38 556	52 000	103 200	174 130	207 040	212 800	215 790	216 980	278 120
		57 330	57 330	57 330	57 330	57 330	57 330	57 330	57 330	96 467
		2661	48 681	139 720	158 920	169 620	174 130	175 490	175 750	247 310
		14 841	23 095	55 305	117 300	163 780	176 860	183 340	185 790	247 310
38	a_2	35 061	35 061	35 061	35 061	35 061	35 061	35 061	35 061	74 197
		27	27	27	27	1297	9999	41 878	100 030	210 420
		10 827	10 827	10 827	10 865	11 278	13 851	23 845	48 649	210 420
		32 427	32 427	32 427	32 427	32 427	32 427	32 427	32 427	71 563
56	a_1	13.5	13.5	13.5	13.5	13.5	13.5	9548	42 197	141 680
		10 814	10 814	10 814	10 814	10 814	11 100	13 223	21 432	141 680
		32 414	32 414	32 414	32 414	32 414	32 414	32 414	32 414	71 550
		0	0	0	0	0	0	0	0	71 550

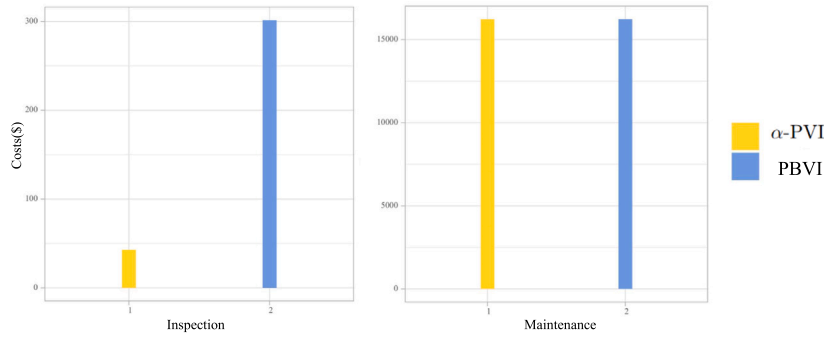


Fig. 7. The comparison of inspection and maintenance costs between the α -PVI and PBVI algorithms.

may influence the degradation, and the signal provides an opportunity to refine the information about the underlying degradation. In other words, the RUL prediction is affected by the current action and signals. Hence, the estimated RUL prediction is expressed as

$$F^{a_l}(t|O_{0:t_d}) = \sum_{i=1}^n F_j(t_d) B_i^{O_{0:t_d}}(t_d) P_{ij}^{a_l} \quad (\text{A.1})$$

where $F_j(t_d)$ represents the RUL when the current state is j . $\square$

Appendix B. The proof of Proposition 2

The inspection timings are predicted based on the RUL distribution by balancing the cost of maintenance and the risk of failure. We set an economic objective subject to the reliability threshold. The economic objective is the minimal cost rate $c(\tau)$ among the successive decision epochs. The predicted inspection interval τ is the decision variable. The reliability threshold R_T is preset and is formulated as

$$\bar{F}^{a_l}(t|O_{0:t_d}) \geq R_T \quad (\text{B.1})$$

where $\bar{F}^{a_l}(t|O_{0:t_d}) = 1 - F^{a_l}(t|O_{0:t_d})$. The value of R_T is set according to experts' knowledge or engineering requirements.

For the age replacement model, it has been well known that we cannot find a finite optimal timing when the failure time has an exponential distribution. To avoid this condition, we implement the concept of deviation cost [34]. The deviation cost is defined between the failure time t and the planned inspection timing τ . The integration range of deviation cost is expressed as Eq. (B.2).

$$\int_0^\infty |\tau - t| dF^{a_l}(t|O_{0:t_d}) = \int_0^\tau F^{a_l}(t|O_{0:t_d}) dt + \int_\tau^\infty \bar{F}^{a_l}(t|O_{0:t_d}) dt \quad (\text{B.2})$$

The expected interval from the current inspection timing to the next inspection timing is

$$\tau \bar{F}^{a_l}(t|O_{0:t_d}) + \int_0^\tau t dF^{a_l}(t|O_{0:t_d}) = \int_0^\tau \bar{F}^{a_l}(t|O_{0:t_d}) dt \quad (\text{B.3})$$

Combining Eqs. (B.2) and (B.3), the expected cost rate during the successive inspection timings is expressed as Eq. (B.4).

$$c(\tau) = \frac{\int_0^\tau c_F \cdot F^{a_l}(t|O_{0:t_d}) dt + \int_\tau^\infty c_L \cdot \bar{F}^{a_l}(t|O_{0:t_d}) dt}{\int_0^\tau \bar{F}^{a_l}(t|O_{0:t_d}) dt} \quad (\text{B.4})$$

where c_F and c_L represent the deviation cost rates after failure and before failure. The relationship between c_F and c_L affects the optimal result. The economic objective can be expressed as

$$t_{d+1} = \arg\min_\tau c(\tau) + t_d \quad (\text{B.5})$$

Then, the predictive inspection timing is optimized as Eq. (B.6) subject to Eq. (B.1).

$$t_{d+1} = t_d + \arg\min_\tau \frac{\int_0^\tau c_F F^{a_l}(t|O_{0:t_d}) dt + \int_\tau^\infty c_L \bar{F}^{a_l}(t|O_{0:t_d}) dt}{\int_0^\tau \bar{F}^{a_l}(t|O_{0:t_d}) dt} \quad (\text{B.6})$$

To compute the optimal τ , we differentiate Eq. (B.4) with respect of τ and set to zero. The formula can be obtained as Eq. (B.7).

$$\int_0^\tau \frac{F^{a_l}(\tau|O_{0:t_d}) - F^{a_l}(t|O_{0:t_d})}{\bar{F}^{a_l}(\tau|O_{0:t_d})} dt = \frac{c_L}{c_F} \mu \quad (\text{B.7})$$

The left is an increasing function of τ from 0 to ∞ . Hence, without constraints, the optimal τ can be obtained. However, in real practice, the reliability threshold should be incorporated. Hence, considering the $c(\tau)$ is a first decreasing and then increasing function of τ , we can

obtain the optimal τ^* satisfies $\tau^* = \min\{\tau_1, \tau_2\}$, where τ_1 and τ_2 satisfy

$$\int_0^{\tau_1} \frac{F^{a_l}(\tau_1 | O_{0:t_d}) - F^{a_l}(t | O_{0:t_d})}{\bar{F}^{a_l}(\tau_1 | O_{0:t_d})} dt = \frac{c_L}{c_F} \mu \quad (\text{B.8})$$

$$\bar{F}^{a_l}(\tau_2 | O_{0:t_d}) = R_T \quad \square \quad (\text{B.9})$$

Appendix C. The proof of Proposition 3

We assume the value function of the α -PVI algorithm can be formatted as

$$V(\mathbf{B}^{O_{0:t_d}}(t_d)) = \min_{\mathcal{L}} \left[\sum_{i=1}^{i=N} \alpha_i^{\mathcal{L}}(t_d) B_i^{O_{0:t_d}}(t_d) \right] \quad (\text{C.1})$$

The Proposition 3 is demonstrated by induction. First, the initial value function follows this format at T as

$$V(\mathbf{B}^{O_{0:T}}(T)) = \sum_i B_i^{O_{0:T}}(T) q_i(T) = \mathbf{B}^{O_{0:T}}(T) \mathbf{q}(T) \quad (\text{C.2})$$

where $\mathbf{q}(T)$ is the corresponding α -vector. Then, the value function at decision epoch t_{d+1} is assumed as following the format of Eq. (C.1). According to Eq. (3), the value function at decision epoch t_{d+1} can be expressed as Eq. (C.3).

$$V(\mathbf{B}^{O_{0:t_d,k}}(t_{d+1})) = \min_{\mathcal{L}} \frac{\left[\sum_j \alpha_j^{\mathcal{L}}(t_{d+1}) \sum_i B_i^{O_{0:t_d}}(t_d) P_{ij}^a(t_{d+1} - t_d) E_{jk} \right]}{\sum_i \sum_j B_i^{O_{0:t_d}}(t_d) P_{ij}^a(t_{d+1} - t_d) E_{jk}} \quad (\text{C.3})$$

The $\mathcal{L}$ is a function of the current action a_l and the signal k . It can be expressed as

$$\mathcal{L} = \mathcal{L}(a_l, k) \quad (\text{C.4})$$

Based on Eq. (11), the value function at decision epoch t_d can be expressed as Eq. (C.5).

$$\begin{aligned} V(\mathbf{B}^{O_{0:t_d}}(t_d)) \\ = \min_{a_l} \left[\sum_i B_i^{O_{0:t_d}}(t_d) \left[q_i^{a_l} + \sum_{k,j} P_{ij}^{a_l}(t_{d+1} - t_d) E_{jk} \alpha_j^{\mathcal{L}(a_l,k)}(t_{d+1}) \right] \right] \end{aligned} \quad (\text{C.5})$$

Hence, the optimal value function is piecewise linear and convex when the decision epochs are non-periodic. $\square$

Appendix D. The proof of Proposition 4

We introduce the concept of Stochastic Increasing as preliminary knowledge. For two vectors $\mathbf{X} = (x_1, x_2, \dots, x_n)$ and $\mathbf{Y} = (y_1, y_2, \dots, y_n)$, if it holds that $\sum_{i=k}^n x_i \geq \sum_{i=k}^n y_i$ for any $0 \leq k \leq n$, then $\mathbf{X}$ is stochastic increasing than $\mathbf{Y}$. Then we apply this concept to demonstrate Proposition 4. We assume the following practical order of constants: $c_1 < c_I < c_2 < c_3$. To begin, we show that a control limit exists between the implementations of a_1 and a_2 . Value function of Eq. (9) under a_1 and a_2 can be expressed as

$$\begin{aligned} V^{a_1}(\mathbf{B}^{O_{0:t_d}}(t_d)) &= c_I + c_1 + \\ &\sum_{i=1}^{i=n} B_i^{O_{0:t_d}}(t_d) \sum_{j=1}^{j=n} P_{ij}^{a_1}(t_{d+1} - t_d) \sum_k E_{jk} V(\mathbf{B}^{O_{0:t_d,k}}(t_{d+1})) \end{aligned} \quad (\text{D.1})$$

$$\begin{aligned} V^{a_2}(\mathbf{B}^{O_{0:t_d}}(t_d)) &= c_I + c_2 + \\ &\sum_{i=1}^{i=n} B_i^{O_{0:t_d}}(t_d) \sum_{j=1}^{j=n} P_{ij}^{a_2}(t_{d+1} - t_d) \sum_k E_{jk} V(\mathbf{B}^{O_{0:t_d,k}}(t_{d+1})) \end{aligned} \quad (\text{D.2})$$

The order of Eqs. (D.1) and (D.2) are assessed under two distinct limit conditions. The first scenario corresponds to systems in their

initial state, where it is typically the case that $V^{a_1} < V^{a_2}$ often exists. The second scenario describes system failure, where the inequality $V^{a_1} > V^{a_2}$ holds. To explore the monotonicity of $V^{a_1}(\mathbf{B}^{O_{0:t_d}}(t_d))$ with respect to $\mathbf{B}^{O_{0:t_d}}(t_d)$, we employ an inductive approach. $V(\mathbf{B}^{O_{0:T}}(T))$ is initialized as an increasing function with respect to stochastic increasing of $\mathbf{B}^{O_{0:T}}(T)$. $\mathbf{B}_1^{O_{0:t_d}}(t_d)$ is assumed as stochastic increasing than $\mathbf{B}_2^{O_{0:t_d}}(t_d)$, implying $V^{a_1}(\mathbf{B}_1^{O_{0:t_d}}(t_d)) \geq V^{a_1}(\mathbf{B}_2^{O_{0:t_d}}(t_d))$. Similarly, the monotonicity of $V^{a_2}(\mathbf{B}^{O_{0:t_d}}(t_d))$ is demonstrated. Consequently, a control threshold exists between the implementations of a_1 and a_2 . By analogous logic, we can also establish that control limits exist for the implementation of actions of a_1 , a_2 , and a_3 . $\square$

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